

SHAIFALI M. ARORA

Summary:

- Ph. D. in the area of Information Technology from GGSIPU, New Delhi on the topic Design Of Fast Motion Estimation Algorithms For Video Encoder Using New Zero Motion Prejudgment And Initial Search Center Prediction Techniques.
- Around 19 years of teaching experience.
- Currently working as Associate Professor in Electronics & Communications Engineering Department, Maharaja Surajmal Institute of Technology, Janakpuri, New Delhi.
- Around 41 papers in Journals and Conferences out of which 19 are in reputed International Journals.
- Awarded with **Certificate of Excellence in Research**, 2019 by Surajmal Memorial Education Society in 2019.
- **Awarded with Certificate of Appreciation** for achieving excellent results from Surajmal Memorial Education Society in 2016.
- **Awarded with Certificate of Appreciation** from Director, MSIT for delivering expert lecture on “Deep Learning and Text Analysis Using Deep Neural Networks” at MSIT in 2018.
- **Awarded with Certificate of Appreciation** from North Cap university for successfully organizing a **Session** in International conference on Computational Intelligence and Data Science.(ICCIDS 2018)
- **Awarded with FACE of the QUARTER** (Best Teacher award) award in Jan, 2008 by honorable Director of Chitkara Trust for my Punctuality, Excellent feedback from students, Design of special experiments in Labs, Taking initiative and successfully completing various conferences and workshops. In every quarter of the year, this special award is given to one of the faculty members associated with “**CHITKARA TRUST**” for their extra ordinary performance.
- **Awarded with scholarship certificate** from Punjab School Education Board for having 33rd rank in matriculation examination at state level. Also stood third in Patiala district in matriculation examination.
- **Organized** a National Conference on Wireless Networks and Embedded Systems “**WNES-06**” on 28th July, 2006 .

- `Delivered **Expert Lectures** on DSP and Image Processing in various FDPs.
- Lead the online teaching initiative with new tools, technologies, working for socially relevant research, managing people with various administrative roles, and connecting people in classrooms.
- Guided 1 Ph. D. scholar as Co-guide.

Education

Ph.D. from University School of Information and Communications Technology, Guru Gobind Singh Indraprastha University, Delhi on the topic “Design Of Fast Motion Estimation Algorithms For Video Encoder Using New Zero Motion Prejudgment And Initial Search Center Prediction Techniques” in March, 2017 under the guidance of Prof. Navin Rajpal.

M.Tech (Part Time) in Electronics & Comm Engg. from Guru Nanak Dev Engineering College, Ludhiana under Punjab Technical University with an aggregate of 75% marks.

B.Tech in Electronics & Comm Engg. from Guru Nanak Dev University, Amritsar With Honours(74%) in May 2002.

Matriculation(10th) from C.M. Model Senior Secondary School, (Rajpura) under Punjab School Education Board with Distinction(90%) in 1996.

Senior Secondary(10+2) from ICL Public school(Rajpura) under Central Board Of Secondary Education, New Delhi with 82% in 1998.

Area of Specialization

Digital Image and Video Processing, Machine Learning, Digital Signal Processing, Network Analysis & synthesis, Microcontrollers, Signal & Systems

Professional Experience

1. Associate Professor with **Maharaja Surajmal Institute of Technology, Janakpuri**, New Delhi from June,2017 till date.
 - Joined this institute as lecturer in 2009
 - Taking B.Tech classes
2. Worked as **Senior Lecturer** in Electronics & Communication Engg Dept. at **Chitkara Institute of Engineering And Technology, Rajpura, Punjab** from **1st August, 2007** to **15th Jan, 2009**. Earlier worked as Lecturer in above college from

1st August, 2005 to 31ST July 2005. Then got promotion as **Senior Lecturer.**

3. **Worked** as Lecturer in **Electronics & Communication Engg** dept. at **Guru Nanak Dev Engg. College, Ludhiana** from 2nd August, 2004 to 31st July, 2005.
4. **Worked** as Lecturer in **Electronics & Communication Engg** dept. at **Shaheed Udham Singh College Of Engg. & Tech., Tangori, Mohali** from June 2002 to July 2004.

Professional Membership

Registered as a **Life Time Member(LM-43690)** with **Indian Society of Technical Education (I.S.T.E) NEW DELHI.**

Research Contribution:

- Published 19 papers in various National & International Journals.(Annexure 1)
- Published 22 papers in various National & International Conferences(Annexure 2)
- Attended FDP/Workshops/Seminars/Conferences

Book/Book Chapters Published

1. Optimizing Transition Band Samples in FIR Filters By Using Particle Swarm Optimization, AkiNik Publications, ISBN: 978-93-85895-45-6, 2016
2. Block-Based Motion Estimation: Concepts and Challenges, Handbook of Research on Advanced Concepts in Real-Time Image and Video Processing, IGI Global pp 51-76; July 2017
3. ARPS: An Efficient fast block matching algorithm for motion estimation, Lambert Academic Publishing, Germany, ISBN(978-620-3-46496-2)

Professional Service:

Workshops/ FDPs/ Seminars Attended

Attended one week FDP on **Emergency Trends in OFC Technology** organized by NITTTR, Chandigarh (Online) from Oct 19 to Oct 23, 2020

Attended one week FDP on **Machine Learning using Python** at MSIT from Mar 2 to Mar 6, 2020.

Attended one week FDP on **Wireless Propagation** from NITTTR, Chandigarh held at MSIT from Feb 25 to March 1, 2019

Attended two days' workshop on **Antenna Designing and Measurements** held at MSIT from March 1 to March 2, 2019.

Attended one week FDP on **4G & 5G through ICT** from NITTTR, Chandigarh held at MSIT from Sep 17 to Sep 21, 2018.

Attended one week FDP on **Modeling & Simulation using MATLAB** organized by NITTTR, Chandigarh held at MSIT, Delhi from May 21 to 25, 2018

Attended one week FDP on **Classroom Communication** from NITTTR, Chandigarh held at MSIT, Delhi from May 28 to June 01, 2018

Attended one week FDP on **MATLAB and SCI Lab** from NITTTR, Chandigarh held at MSIT from Sep 18 to 22, 2017.

Attended one week FDP on **Deep Learning for Image and Text Analysis with Python** organized by BVCOE, New Delhi from Nov 27 to Dec 1, 2017.

One day FDP on **Design and characteristic of RF Antenna** on 9th Feb 2017 held at MSIT.

Attended one day FDP on **Time Management by Teacher in Class** on 2nd March 2015 held at MSIT.

Attended One day FDP on **Train the Trainer** on 2nd March 2015 held at MSIT.

Attended One day FDP on **Ethics and Responsibilities of Teachers Towards Students and the Institute** on 2nd March 2015 held at MSIT.

Attended One day FDP on **Virtual Reality** on 28th February 2015 held at MSIT.

Attended One day FDP on **Digital Design Techniques** on 28th February 2015 held at MSIT.

Attended Three day FDP on **Digital Design using Verilog HDL** from 17th Feb 2014 to 19th Feb 2014 at MSIT.

Attended two days FDP on **Embedded Systems Design using ATMEL XMEGA XPLD A1 and ARM Based SAM4L** on July 18, 2013 to July 19, 2013 organized by ATMEL University at IIIT Delhi.

Attended two days FDP on **Intelligent Tools for Various Engineering Applications** in MSIT on 12th & 13th April 2012

Attended two days FDP on **Embedded Systems using ARM7**; Feb 2, 2012- Feb 3, 2012 at MSIT

Attended 1 day **EFY Expo held at Pragati Maidan** on 22nd February 2014.

Attended One Day Workshop **How to get Published in Research Journal** on Sep 11, 2013 at GGSIPU.

Attended 1-day workshop organized by NPTEL at MSIT, Feb 2, 2013.

Attended one day Seminar on **Designing and Implementing Signal Processing, Video Processing and Communication Systems using MATLAB and Simulink** by Mathworks at Le Meridian on 17th Oct. 2013

Attended one day seminar on **MATLAB and SIMULINK Academic tour** on Sep 24 , 2013, at Hotel Shangri La's Eros Hotel, New Delhi.

Attended one week FDP on **DSP & Applications** at IGIT, New Delhi; June 20,2011 to June 24, 2011

Attended one week FDP on **ANN, FUZZY –LOGIC & Other Soft Computing Techniques** at NITTTR, Chd. from Aug 16, 2010 to Aug 20, 2010

Attended a two day CEEP course on **Advanced Fuzzy Logic Technology & Its Applications** held at Electrical & Instrumentation Department at Thapar Institute of engineering & Technology, Patiala from Oct 25, 2006 to Oct 26,2006.

Attended one week FDP on **DSP and its Applications** organized by Electronics & Electrical Engineering Department at Punjab Engineering College from 4th July, 2006 to 7th July, 2006

Attended a two week AICTE sponsored FDP on **Induction Training Program For Faculty of Engineering Colleges** organized by NITTTR, Chandigarh at Chitkara Institute of Engineering & Technology campus from Feb 27, 2006 to March 10, 2006.

Attended one week FDP on **Curriculum Processes** organized by NITTTR, Chandigarh at SUS Engineering College, Tangori, Mohali from Sep 23, 2003 to Sep 26, 2003.

Workshops/FDPs Organized

Organized a National Conference on Wireless Networks and Embedded Systems “**WNES-06**” on 28th July, 2006 .

Delivered Expert **Lecture on DSP** in a week long workshop on “Control System and Digital Signal Processing using MATLAB” for the faculty of various Engineering and Polytechnic colleges from 2nd Jan to 6th Jan, 2007

Organized a week long workshop on “Embedded System Design using Microcontrollers” for the faculty of various Engineering and polytechnic colleges from 18th Dec to 25th Dec , 2007.

Personnel Particulars

Marital Status: Female- Married, Two Kids
Father's Name : Sh. Bharat Bhushan Madan
Mother's Name : Smt. Nirmal Madan
Husband's Name : Mr. Pankaj Arora
Date of Birth : 23-04-1981
Languages Known : Hindi, English, Punjabi
Add for correspondence: Block No.-4, House No.-32
Ground Floor, Moti Nagar
New Delhi-110015

(Dr Shaifali M. Arora)

References:

- 1) Professor Navin Rajpal
University School of Information & Communication Technology
Guru Gobind Singh Indra Prastha University , Delhi
e-mail navin.rajpai@ipu.ac.in

2)

Annexure 1

Journal Publication:

1. Kadian, Poonam, **Shaifali M. Arora**, and Nidhi Arora.2021 "Robust Digital Watermarking Techniques for Copyright Protection of Digital Data: A Survey." *Wireless Personal Communications* 2021: 1-25.
2. **Arora, S.M.**, Rajpal, N. and Khanna, K., 2019. A new approach with enhanced accuracy in zero motion prejudgment for motion estimation in real-time applications. *Journal of Real-Time Image Processing*, 16,4, 989-1005
3. Kadian, Poonam, Nidhi Arora, and **Shaifali M. Arora**. 2019, "A Highly Secure and Robust Copyright Protection Method for Grayscale Images using DWT-SVD.", *International Journal of Recent Technology and Engineering*
4. Kadian, Poonam, Nidhi Arora, and **Shaifali M. Arora**. 2019, "Role of Scaling Factor in Robust Digital Watermarking" *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, Vol :8,11, 1658-1669

5. Pareek, A.; **Arora, S. M.** 2017. Breast cancer detection techniques using medical image processing. *International Journal of Multidisciplinary Education and Research*, 2, 3, 79-82
6. Poonam. and **Arora, S.M.**, 2017, Digital Watermarking: An Introduction, *International Journal of Applied Research* 3 (4), 128-131
7. **Arora, S.M.**, Rajpal, N., Purwar, R., 2017. A new fast motion estimation algorithm using adaptive size diamond pattern search with early search termination., *International Journal of Computational Vision and Robotics*
8. **Arora, S.M.**, Rajpal, N., Khanna, K., 2017. A Novel Hybrid Approach for Fast Block Based Motion Estimation. Accepted for publication in *International Journal of Interactive Multimedia and Artificial Intelligence*.
9. **Arora, S.M.**, Rajpal, N., Khanna, K. and Purwar, R., 2016. Improved accuracy in initial search center prediction to fasten motion estimation in h. 264/AVC. *IETE Journal of Research*, 62(6), pp.842-851.
10. **Arora, S.M.**, Rajpal, N., Khanna, K. and Purwar, R., 2016. Comparative analysis of various search center selection methods for fast motion estimation. *IJAR*, 2(4), pp.183-186.
11. **Arora, S.M.**, 2016, Fast Motion Estimation by Early Search Termination. *International Journal of Academic Research and Development*, 7(1), pp 51-62
12. **Arora, S.M.**, 2016, Trends and Issues in developing fast motion estimation algorithms. *International Journal of Applied Research*, 2(9), pp 607-613
13. **Arora, S.M.**, 2016, Variable search windows for slow, medium and fast motion sequences. *International Journal of Research in Advanced Engineering and Technology*, 2(3), pp 165-167
14. **Arora, S.M.**, 2016, Fast motion estimation algorithms. *International Journal of Multidisciplinary Research and Development*, 3(3), pp 165-167
15. Khanna, K. and **Arora, S.M.**, 2016. Ant Colony Optimization towards Image Processing. *Indian Journal of Science and Technology*, 9(48).

16. Kavita, K., Navin, R. and **Madan, A.S.**, 2016. Piecewise feature extraction and artificial neural networks: an approach towards curve reconstruction. *Indian Journal of Science and Technology*, 9(28).
17. Khanna, K., Rajpal, N., Dureja, A. and **Arora, S.M.**, 2016. Surface Reconstruction using Feature based Approach with Radial Basis Function Neural Network. *IJCTA*, 9(19), pp. 9239-925
18. Batra, B., Singh, S., Sharma, J. and **Arora, S.M.**, 2016, Computational analysis of edge detection operators. *International Journal of Applied Research*, 2(11)
19. Singh, A.K., Roy, S., Sodhi, S., Goel, D. and **Arora, S.M.**, 2014. SmartX Virtuality: A Smarter way to Interact Virtually with Physical Objects. *International Journal of Scientific and Research Publications*, 4(6).

Annexure 2

Conference Publications:

1. Mishti Gautam, Shaifali M. Arora, 2021, Smart Monitoring and Fault Detection of Sewer System (SMFD) , National Conference on Advances in Metrology, 2021
2. Kadian, P., Arora, N. and **Arora, S.M.**, 2020, Robust Watermarking Schemes for Copyright Protection of Digital Data: A Survey , International Conference on Emerging Technologies in Data Science in data Mining and Information Security, IEM, Kolkata
3. Kadian, P., Arora, N. and **Arora, S.M.**, 2019, March. Performance Evaluation of Robust Watermarking Using DWT-SVD and RDWT-SVD. In 2019 6th International Conference on Signal Processing and Integrated Networks (SPIN) (pp. 987-991). IEEE.
4. **Arora, S. M.** , P. Kadian 2018. A DWT-SVD based robust digital watermarking for digital images. *Procedia computer science*, 132, 1441-1448.
5. **Arora, S.M.**, Khanna, K., 2016, November. **Dynamic Two Level Threshold Estimation For Zero Motion Prejudgment: A Step Towards Fast Motion Estimation.** *Advances in Computing and Data Sciences (ICACDS)*, 2016. First International conference.
6. **Arora, S.M.**, Rajpal, N. and Purwar, R., 2015, February. Dynamic pattern search algorithm with zero motion prejudgment for fast motion estimation. In *Advanced*

- Computing & Communication Technologies (ACCT), 2015 Fifth International Conference on* (pp. 138-142). IEEE.
7. **Arora, S.M.**, Rajpal, N. and Purwar, R., 2014, September. New dynamic pattern search based fast motion estimation algorithm. In *Advances in Computing, Communications and Informatics (ICACCI, 2014 International Conference on* (pp. 979-985). IEEE.
 8. **Arora, S.M.** and Rajpal, N., 2014, September. Survey of fast block motion estimation algorithms. In *Advances in Computing, Communications and Informatics (ICACCI, 2014 International Conference on* (pp. 2022-2026). IEEE.
 9. **Arora, S.M.** and Rajpal, N., 2014, February. Comparative analysis of motion estimation algorithms on slow, medium and fast video sequences. In *Optimization, Reliability, and Information Technology (ICROIT), 2014 International Conference on* (pp. 422-427). IEEE.
 10. Chatrath, J., Gupta, P., Ahuja, P., Goel, A. and **Arora, S.M.**, 2014, February. Real time human face detection and tracking. In *Signal Processing and Integrated Networks (SPIN), 2014 International Conference on* (pp. 705-710). IEEE.
 11. Gupta, S. and **Arora, S.M.**, 2011. Motion Estimation Method between Diamond Search and Three Step Search. In *Proceedings of National Conference On Web and Knowledge Based Systems (WKBS) Emerging Trends and Perspectives* (p. 33).
 12. **Madan, S.**, 2010, October. Optimizing transition band samples in FIR filters by using particle swarm optimization. *Advances in Digital Signal Processing (NCADSP)*, 2010. In national conference on (pp 203-209).
 13. **Madan, S.**, 2006, November. Microprocessor based automatic room light controller with up/down counter. *Emerging trends in Electrical, Electronics and Communication Engineering (EETEEEC)*, 2006. In National conference on (pp 188-192)
 14. Kaur, S., **Madan, S.**, 2006, November. Noise removal in Medical Images. *Recent Advances in Electronics & Communication Technology (RAECS)*, 2006, National Conference on (pp 486-489).
 15. **Madan, S.**, Kaur, S. 2006, November. Simulation of MFCC based Speaker

- Recognition System. Recent Advances in Electronics & Communication Technology (RAECS), 2006 National Conference on (pp 449-452).
16. **Madan, S.** Kaur, S., 2006 , July. Implementation of MFCC Based Speaker Recognition System on MATLAB. Wireless Networks & Embedded Systems, (WNES), 2006, National Conference on (pp 50-53)
 17. Kaur , S., **Madan, S.**, 2006, February. Image segmentation using MATLAB. Electronic Circuits & Communication Systems(ECCS), 2006, National Conference on (pp132-135).
 18. **Madan, S.**, Kaur, S., 2006, February. Image compression using neural Networks. Electronic Circuits & Communication Systems (ECCS), 2006, National Conference on (pp 127-131).
 19. Bhambri, P., Rattan, M., **Madan, S.**, 2005, February. Institution for a Sustainable Civilization: Negotiating Changes in a Technological Culture. In Technical Education in a Globalized Environment (TEGE), 2005, National Conference on (pp 7-9).
 20. **Madan, S.**, Rattan, M., Bhambri, P. 2005, March. Information Retrieval using soft Computing Techniques. In Bioinformatics Computing (NCBC), 2005, National Conference on (pp 159-163).
 21. Rattan, M., **Madan, S.**, Bhambri , P., 2004, December. Smart antenna Technology for Mobile Communication. In Issues & Trends in Wireless Networks (ITWN), 2004 National Conference on (pp 91-94).

Kaur, S., **Madan, S.**, 2004, November. Implementation of ICT Enabled education. In ICT ENABLED EDUCATION (ICTEE), 2004 National Seminar.